

Table 2: Performance comparison of different approaches in the global setting for long-form audio, reported in terms of WDER (%) and cpWER (%). “Without SPC Update” refers to not updating the speakers already stored in the Speaker Prompt Cache.

System	Strategy to Maintain Global Speaker Consistency	Streaming Chunks	SPC Update	CH109 Test		Fisher Test	
				WDER	cpWER	WDER	cpWER
Non-streaming Inference							
DiarizationLM (Llama 3) [23]	Independent ASR & Diarization with LLM Post Processing	-	-	6.66	23.57	3.28	18.37
DiarizationLM (PaLM 2) [23]		-	-	4.25	20.22	2.37	16.93
JEDIS-LLM	Offline Chunk Inference + Global Clustering	-	-	2.48	19.03	2.06	15.03
Chunk-wise Streaming Inference							
JEDIS-LLM	Streaming Inference with Speaker Prompt Cache (SPC)	Oracle Chunks	✗	2.09	18.58	2.51	16.40
			✓	1.73	18.20	2.05	15.88
		VAD Chunks	✗	2.62	19.32	2.95	17.37
			✓	2.54	19.09	2.35	16.60

Table 3: Comparison of streaming inference with and without speaker profiles on the CH109 test set for long-form audio in the global setting, showing cpWER (%), SA-WER (%), and their difference (Δ), where Δ indicates how strictly predicted speaker IDs match the reference.

Streaming Chunks	Speaker Profiles	cpWER	SA-WER	Δ
Oracle Chunks	✗	18.20	25.98	7.78
	✓	17.91	19.98	2.07
VAD Chunks	✗	19.09	30.79	11.7
	✓	19.18	21.94	2.76

matches predicted speaker IDs with the reference without finding best permutation, providing a stricter measure than cpWER.

4. EXPERIMENTAL RESULTS

4.1. Evaluation on Local Setting for Short Audio

Table 1 presents the performance comparison in the local setting (audio clips under 20 seconds) using non-streaming inference. Compared with previous strong baselines (Sortformer [15] and MetaCat [16]), our JEDIS-LLM achieves significantly better cpWER, demonstrating both the advantages of Speech-LLMs for joint ASR and diarization and the effectiveness of our proposed method.

The ablation study further shows that speaker supervision on the encoder is crucial for diarization, as removing it leads to higher WDER. Frame-level speaker supervision [5, 15] improves diarization performance compared to no speaker supervision, but degrades cpWER, indicating that frame-level loss negatively affects the encoder’s ASR capability. Using word-level speaker-attributed transcription as the LLM objective yields reasonable WDER performance, particularly on AMI where speaker turns are frequent, but its cpWER is still worse than that of segment-level transcription. This is because word-level speaker-attributed transcription introduces excessive speaker label splits, disrupting context. Overall, combining segment-level transcription for the LLM with word-level speaker supervision for the encoder yields the best performance.

4.2. Evaluation on Global Setting for Long-form Audio

Table 2 presents the performance comparison in the global setting for long-form audio. Previous work, DiarizationLM [23], aligns independent ASR and speaker diarization results and subsequently employs a finetuned LLM for post-processing. We evaluate our model under both offline and chunk-wise streaming inference. For offline inference, long-form audio is segmented into chunks under 20 seconds based on oracle sentence boundaries, followed by offline inference. The word-timing model and global clustering are then applied to produce global results. For chunk-wise streaming inference, we

employ the Speaker Prompt Cache (SPC) to maintain speaker consistency across chunks.

The results show that our proposed streaming JEDIS-LLM produces substantially better performance, significantly outperforming DiarizationLM using either oracle or VAD chunks, despite the latter being a cascaded system using offline post-processing. As expected, enabling the SPC update mechanism improves performance by refreshing the cache with higher-quality speaker prompts. Moreover, streaming inference surpasses the “Offline Chunk Inference + Global Clustering” baseline on the CH109 test set and achieves the best WDER on the Fisher test set when using oracle chunks. These results demonstrate the effectiveness of SPC and its update strategy.

4.3. Performance of Speaker Profiles Integration

Table 3 presents the results of streaming inference with and without speaker profiles on long-form audio in the global setting, evaluated by cpWER (%), SA-WER (%), and their difference (Δ). Without profiles, reference speaker IDs are assigned by order of appearance (speaker1, speaker2, ...), whereas with profiles they follow the concatenation order of the given profiles. The results show that incorporating profiles narrows the gap between SA-WER and cpWER, indicating that fixed, manually segmented profiles align predicted IDs with reference speakers more effectively than an on-the-fly automatically updated SPC. In addition, profiles allow direct mapping from predicted IDs to real speaker names, rather than index labels, which is especially valuable for real-world applications.

5. CONCLUSION

In this work, we propose an end-to-end Speech-LLM for joint ASR and speaker diarization, trained only on short audio segments under 20 seconds, yet capable of performing chunk-wise streaming inference on long-form audio without additional training. We introduce a speaker prompt cache with an on-the-fly update mechanism, which enables chunk-wise streaming inference while preserving speaker consistency across chunks. Furthermore, replacing the speaker prompt cache with manually defined high-quality utterances allows seamless integration with speaker profiles. In addition, incorporating word-level speaker supervision into the speech encoder during training enhances the model’s diarization capability. Experimental results show that our approach outperforms strong baselines, including Sortformer and MetaCat in the local diarization setting (up to 20 seconds), and DiarizationLM in the global setting for long-form audio, while remaining fully end-to-end and streamable, in contrast to DiarizationLM’s cascaded offline pipeline. To the best of our knowledge, this is the first work to enable streamable joint ASR and diarization on long audio using a Speech-LLM trained only on short audio, achieving state-of-the-art performance.

6. REFERENCES

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7. APPENDIX

7.1. Detail of Training Data

The statistics of the training dataset are shown in Table 4.

Table 4: Statistics of the training data sets.

Dataset	#sessions	#speakers per session	Duration (hours)
AMI train&dev	155	4	90
ICSI all	75	3~11	71
Fisher train	11,527	2	1929
Internal	36,118	2~7	6734
Simulated	21,943	5	964

7.2. Different Chunk and SPC Lengths for Streaming Inference

Beyond the default experimental setting, we further explore different chunk lengths and per-speaker SPC lengths in streaming inference for long-form audio. As shown in Table 5, reducing the SPC length to 3 seconds proves beneficial for ASR. This is because the text in SPC is a hypothesis rather than ground truth; shorter SPCs contain fewer errors, which improves the quality of the initial context for LLM inference. In contrast, reducing the chunk length degrades overall performance, since shorter chunks make it more difficult to find high-quality segments within the chunk for updating the SPC on-the-fly.

Table 5: Performance comparison of different chunk lengths and per-speaker SPC lengths in streaming inference for long-form audio, reported in terms of WDER (%) and cpWER (%).

Streaming Chunks	Chunk Length	SPC Length	CH109 Test		Fisher Test	
			WDER	cpWER	WDER	cpWER
Oracle Chunks	$\leq 10s$	$\leq 5s$	1.73	18.20	2.05	15.88
	$\leq 10s$	$\leq 3s$	2.11	18.43	2.05	15.65
VAD Chunks	$\leq 10s$	$\leq 5s$	2.54	19.09	2.35	16.60
	$\leq 10s$	$\leq 3s$	2.55	18.91	2.20	16.27
	$\leq 5s$	$\leq 5s$	2.85	23.38	2.91	18.35
	$\leq 5s$	$\leq 3s$	2.88	21.46	2.93	18.18
	$\leq 3s$	$\leq 3s$	4.29	25.54	3.83	21.03

7.3. Chain-of-Thought Exploration

Since joint ASR and diarization in multi-talker scenarios is challenging, we explore the use of Chain-of-Thought (CoT) reasoning within the Speech-LLM. Specifically, we prepend a simple reasoning chain before the speaker-attributed transcription to estimate the number of speakers in the input audio, e.g., “*estimated_speaker_number: 3*”. We denote the reasoning chain as C with embedding E^c , and modify Eq. (5) as:

$$\text{Concat}(\hat{C}, \hat{T}) = \text{LLM}(\text{Concat}(E^s, E^p, E^c, E^t)), \quad (7)$$

where the token prediction loss is computed jointly over the reasoning chain and the transcription.

The results on the AMI test set are shown in Table 6. We observe that incorporating CoT yields slightly better WDER and speaker counting accuracy, particularly for utterances involving four speakers, suggesting that CoT helps in estimating larger speaker numbers. When using ground-truth CoT that includes the true number of speakers for inference, WDER is further reduced and speaker counting accuracy approaches 100%, indicating that the reasoning

chain can effectively guide speaker-attributed transcription. However, even when the number of speakers was estimated accurately, the cpWER did not improve, and the WDER improvement was also limited. This suggests that the model did not truly capture the conversational dynamics when provided with the actual number of speakers in the reasoning chain.

Table 6: Performance comparison of our JEDIS-LLM with and without Chain-of-Thought (CoT) on the AMI test set, reported in terms of WDER (%), cpWER (%), and Speaker Counting Accuracy (%). For speaker counting, the number of speakers is taken from the predicted speaker-attributed transcriptions, rather than from the reasoning chain.

CoT Type	WDER	cpWER	Speaker Counting Accuracy				
			1-sp	2-sp	3-sp	4-sp	avg
w/o CoT	6.97	23.13	96.1	84.4	60.0	30.4	68.9
Predicted	6.93	23.43	93.7	82.2	61.0	37.0	69.5
Ground-truth	6.60	23.83	100	100	98.5	98.2	99.2